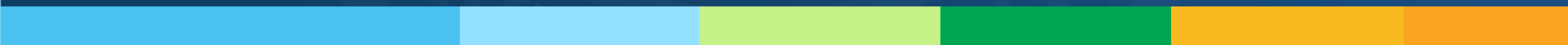




HUMAN BIRTH DEFECTS

Module 1



-
- Langman Medical embryology 15th Edition p130-146
 - Before We Were Born- 9th Edition p303-319

LEARNING OBJECTIVES

1. Define congenital malformations or anomalies.
2. Define teratogens.
3. Give examples of teratogens of different types and the abnormalities they cause.
4. State the principles of teratology.
5. Define numerical and structural chromosomal abnormalities and give examples of each category.
6. Define single gene mutation and give examples of conditions caused by this mechanism.

HUMAN BIRTH DEFECTS

Congenital malformations – are gross structural defects present at birth

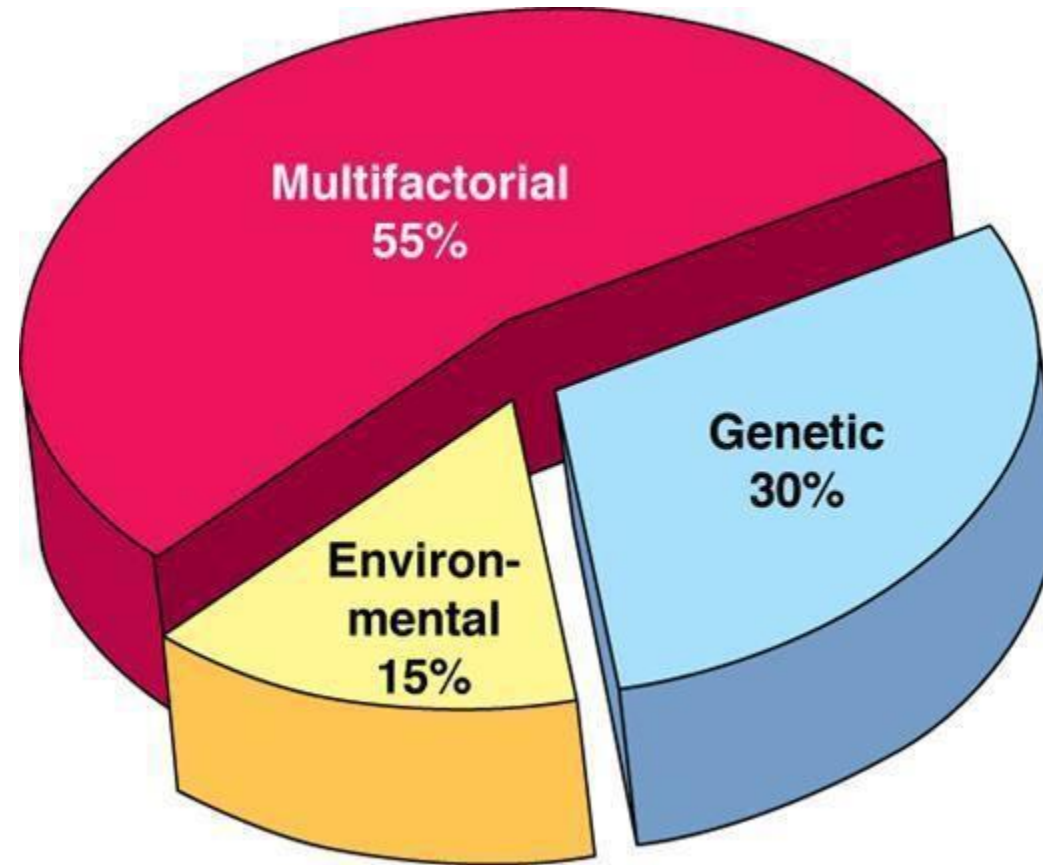
Teratology – study of birth defects and their causes

Teratogen – is any agent that can produce a congenital anomaly or raise the incidence of an anomaly in the population

CONGENITAL MALFORMATIONS

- 2-3% of live newborn infants show 1 or more anomalies at birth; another 2-3% are detected by 5 years of age
- genetic, environmental, or interplay of these factors have been known to cause congenital anomalies; most congenital anomalies result from unknown causes
- time of exposure to factors is crucial

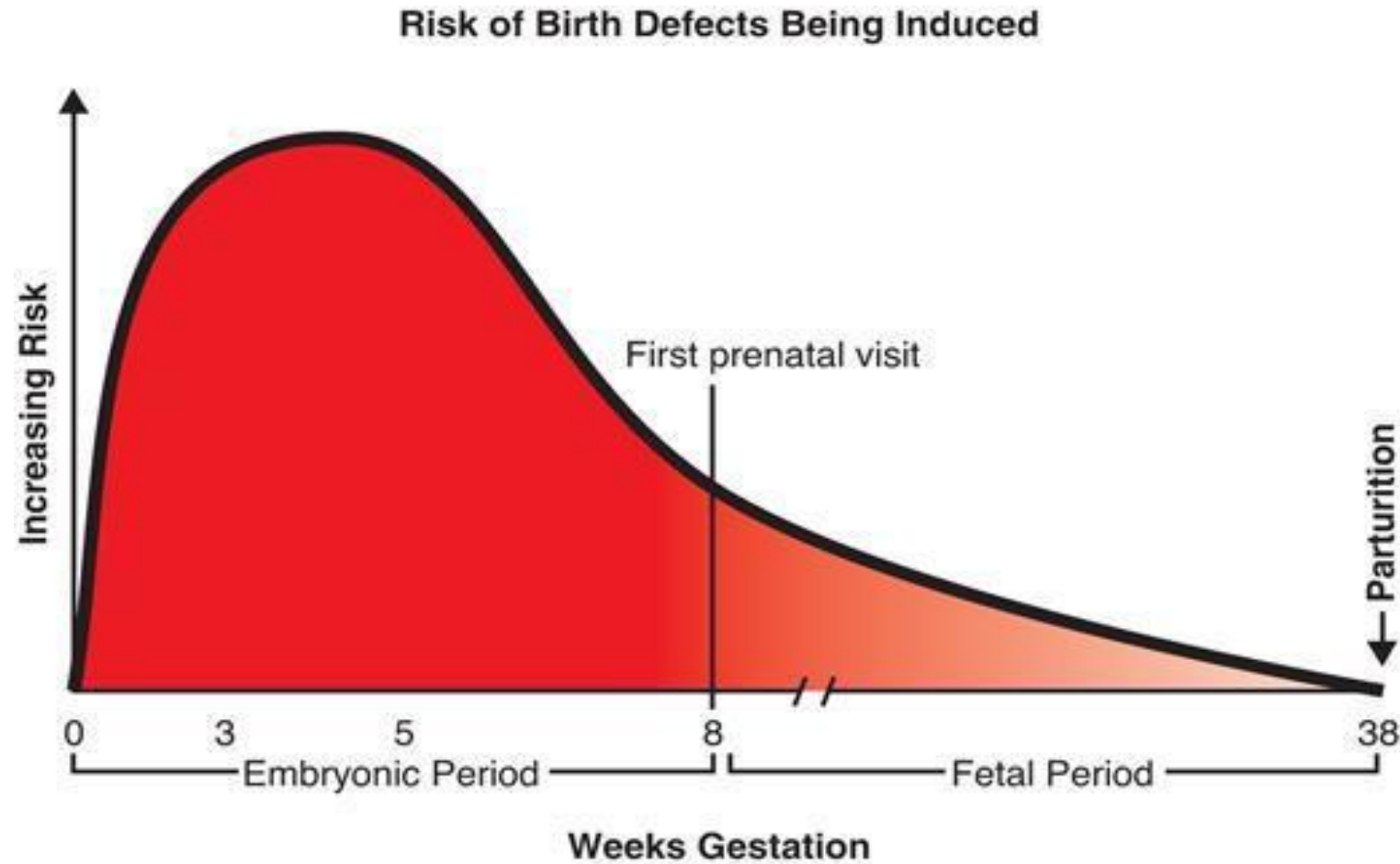
CONGENITAL MALFORMATIONS



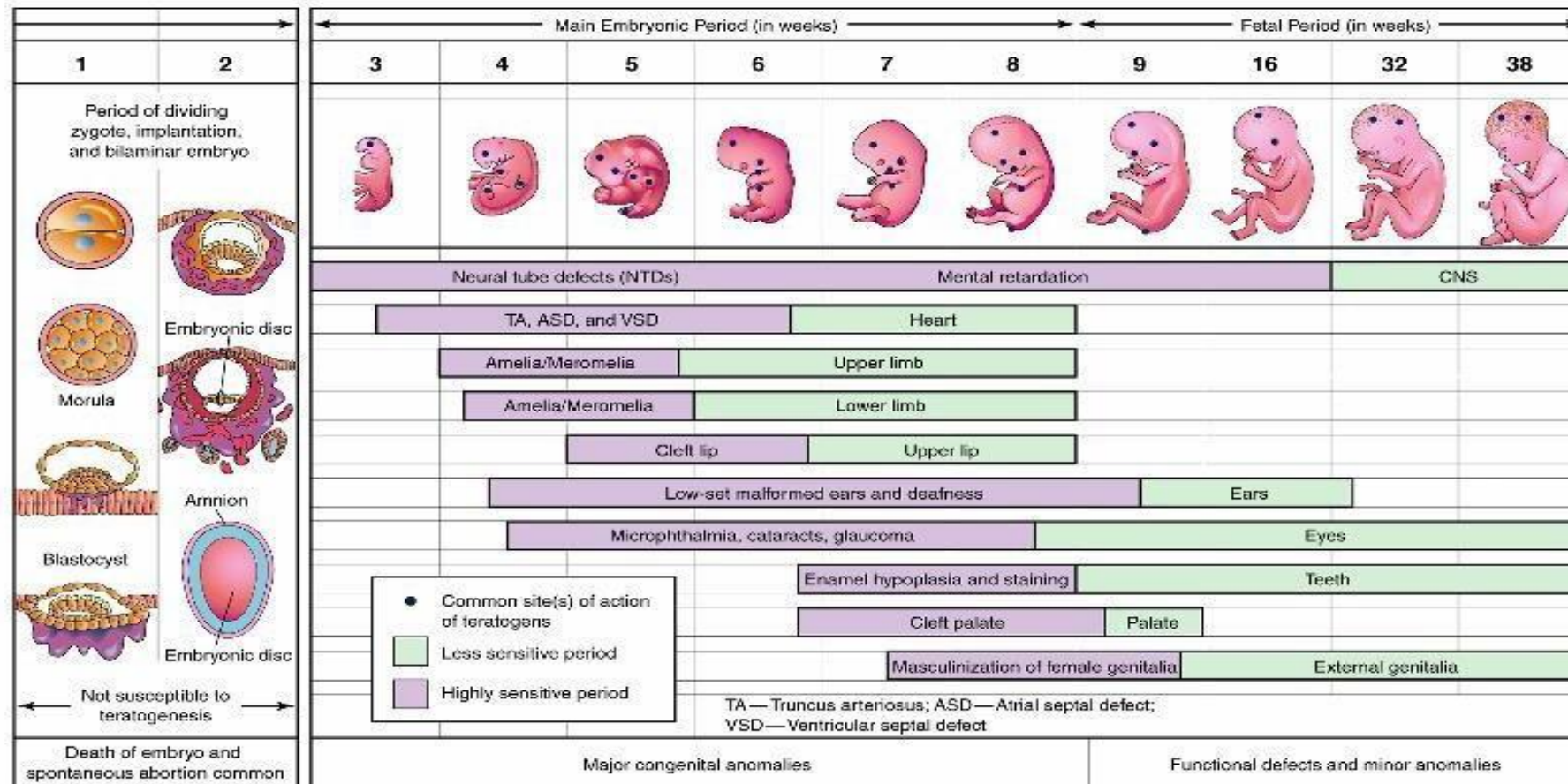
Basic Principles in Teratogenesis

1. Susceptibility to teratogenesis depends on the **genotype of the conceptus** and the manner in which this genetic composition interacts with the environment.
2. Susceptibility to teratogens varies with the **developmental stage at the time of exposure**.
3. Manifestations of abnormal development depend **on dose and duration of exposure** to a teratogen.
4. Teratogens act in specific ways (**mechanisms**) on developing cells and tissues to initiate abnormal embryogenesis (**pathogenesis**). Mechanisms may involve inhibition of a specific biochemical or molecular process; pathogenesis may involve cell death, decreased cell proliferation, or other cellular phenomena.
5. Manifestations of abnormal development are death, malformation, growth retardation, and functional disorders.

CRITICAL PERIOD OF DEVELOPMENT



CRITICAL PERIOD OF DEVELOPMENT



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CONGENITAL MALFORMATIONS

- CHROMOSOMAL/GENETIC FACTORS
- INFECTIOUS AGENTS
- HYPERTHERMIA
- RADIATION
- CHEMICAL AGENTS/DRUGS
- HORMONES
- MATERNAL DISEASE
- HYPOXIA
- ENVIRONMENTAL CHEMICALS

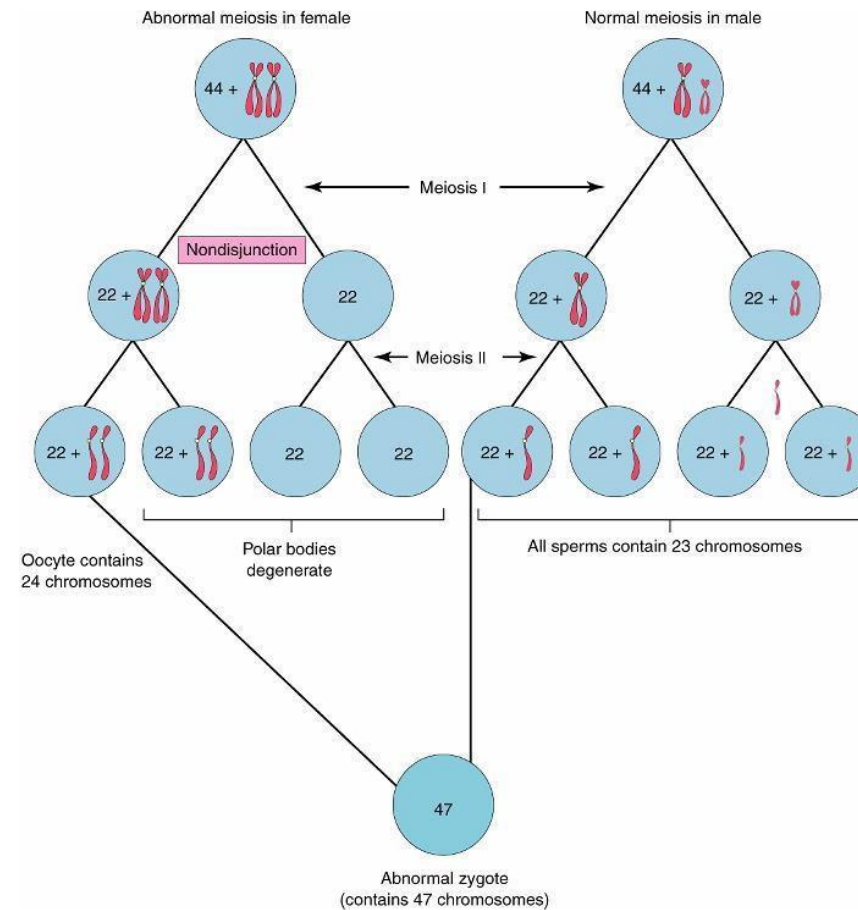
CONGENITAL MALFORMATIONS

CHROMOSOMAL/GENETIC FACTORS

Numerical chromosomal abnormalities due to nondisjunction

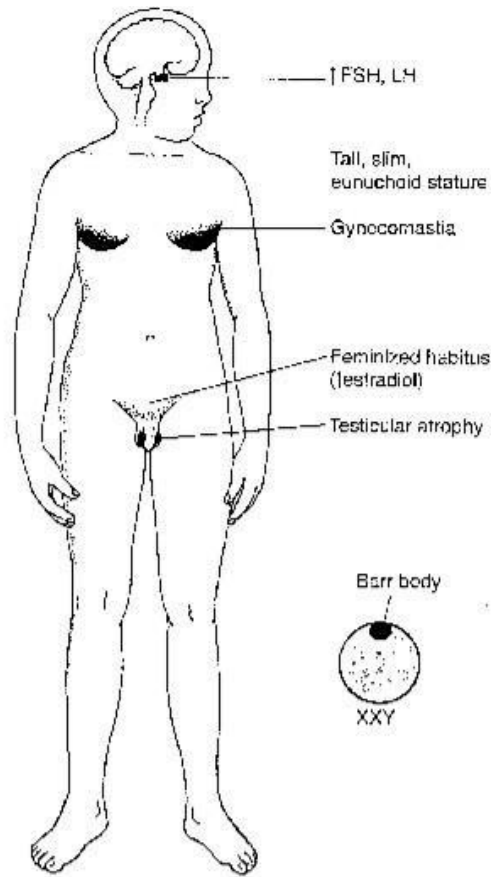
- Turner syndrome (45, X)
- Klinefelter syndrome (47, XXY)
- Trisomy of autosomes (Trisomy 21)
- Trisomy of sex chromosomes
- Mosaicism, Triploidy, Tetraploidy

NONDISJUNCTION



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NONDISJUNCTION



Klinefelter Syndrome (47,XXY)



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NONDISJUNCTION

Turner syndrome

- Short stature
- Webbed neck
- No sexual maturation
- Broad chest



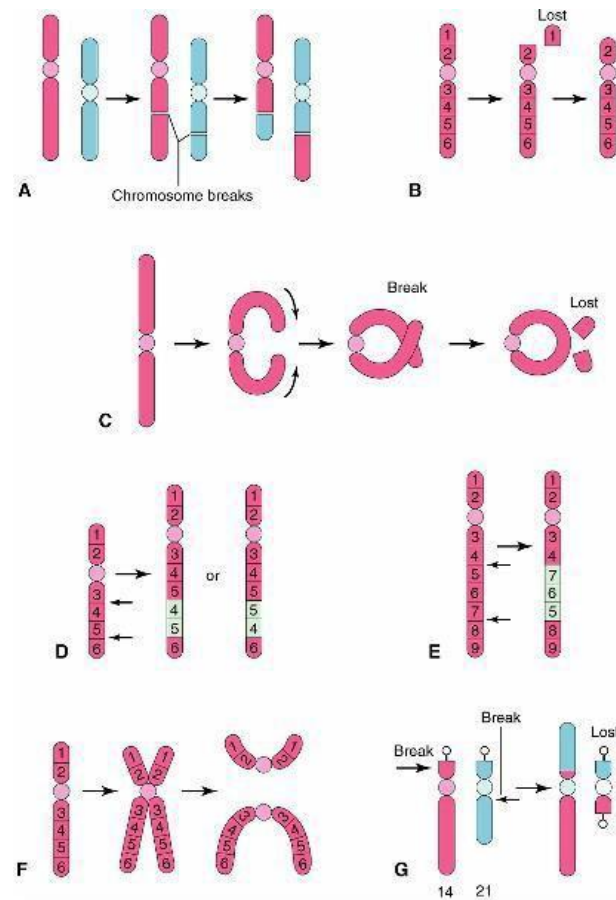
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CONGENITAL MALFORMATIONS

CHROMOSOMAL/GENETIC FACTORS

- Structural chromosome abnormality - caused by translocation or deletion; Ex: cri du chat syndrome (chr. 5)
- Mutant genes - anomalies due to single gene mutations; involves loss or change in the function of a gene; Ex: achondroplasia, fragile X syndrome

TRANSLOCATION OR DELETION

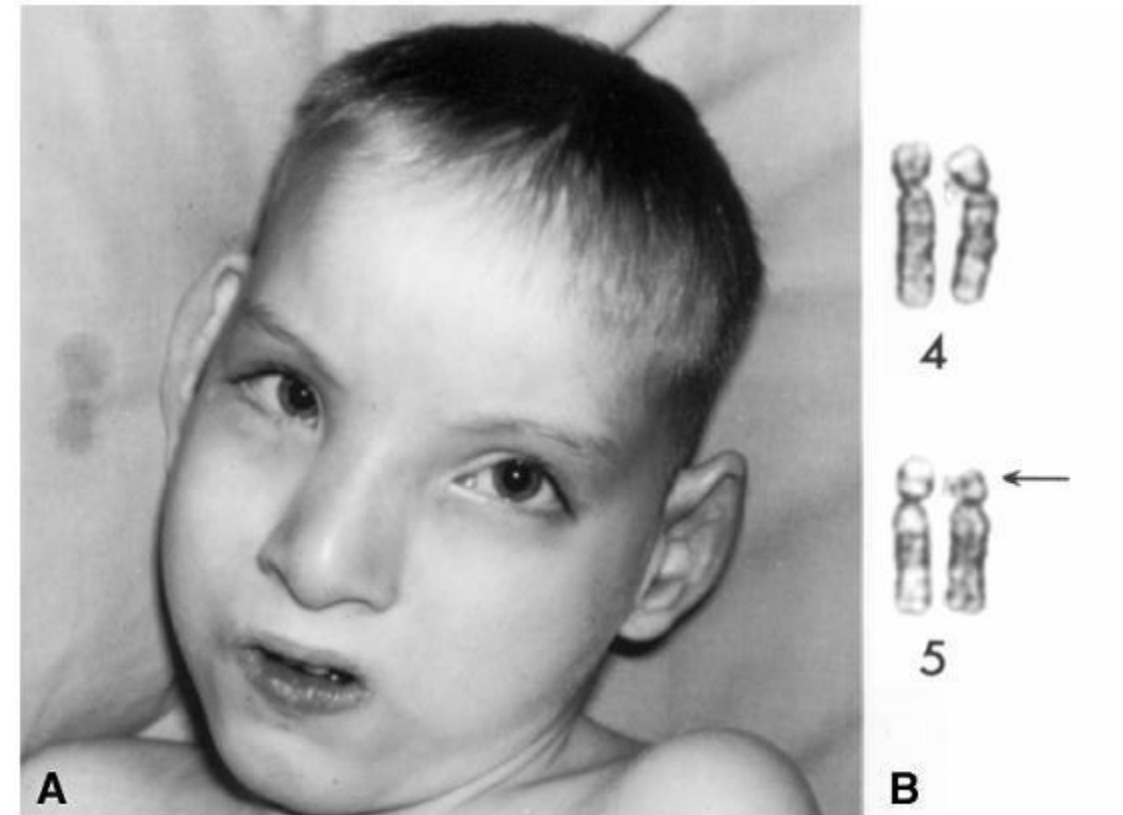


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TRANSLOCATION OR DELETION

Cri-du-chat

Terminal deletion of the short arm of chromosome 5



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TRANSLOCATION OR DELETION

Angelman Syndrome

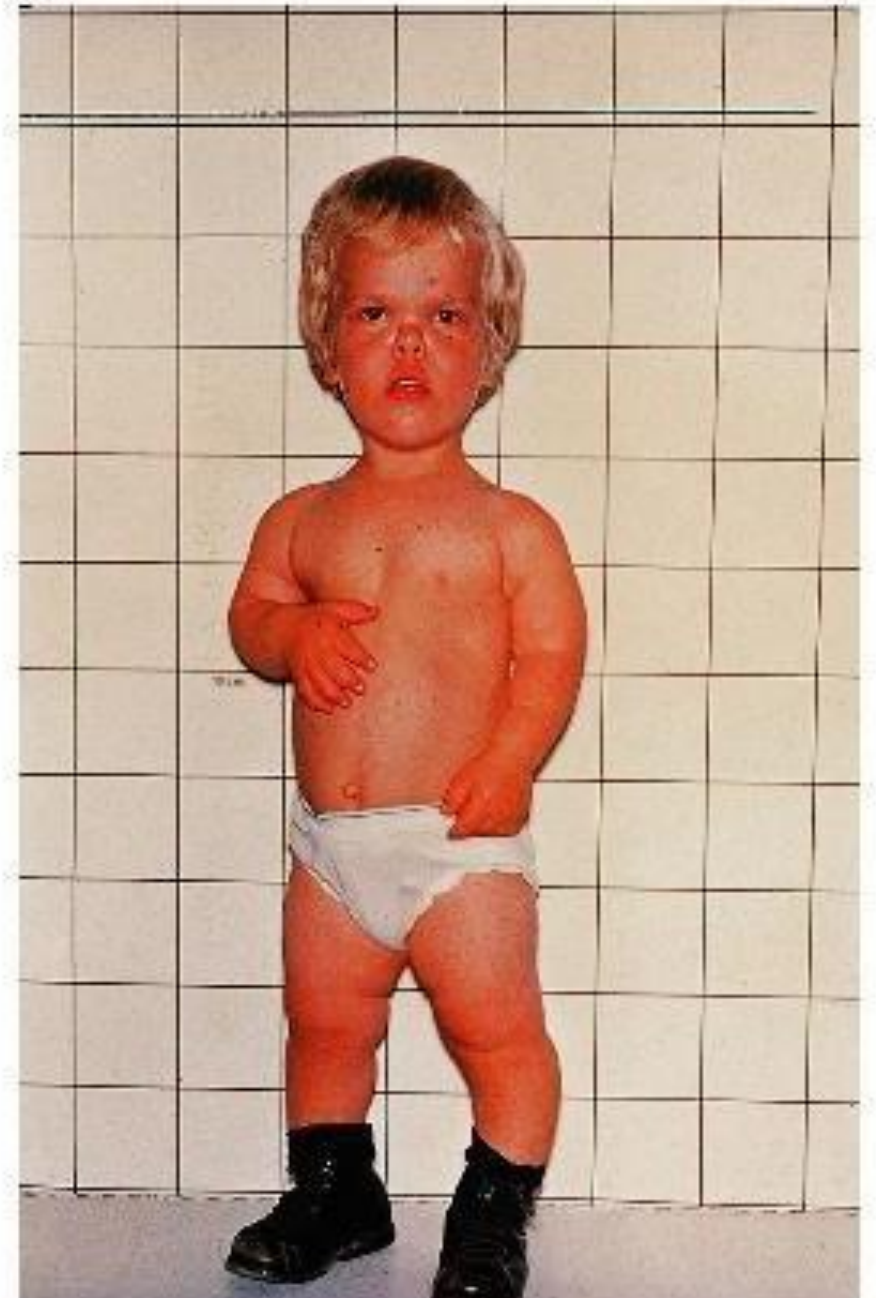
- Deletion of a segment of the maternal chromosome 15

Prader-Willi Syndrome

- Deletion of a segment of the paternal chromosome 15

MUTATION

Achondroplasia – results from a genetic mutation. It is an example of a dominantly inherited congenital anomaly.



MUTATION

Fragile X syndrome – most common inherited cause of moderate mental retardation; it is an X-linked disorder.



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CONGENITAL MALFORMATIONS

INFECTIOUS AGENTS

TORCH syndrome

- a. toxoplasmosis - hydrocephalus, cerebral calcifications, MR
 - b. rubella or German measles
 - c. cytomegalovirus (CMV) - most common viral infection of the human fetus
 - d. herpes simplex virus
1. Zika virus - severe microcephaly
 2. Syphilis - MR, deafness
 3. Varicella or herpes zoster

CONGENITAL MALFORMATIONS

- Infants showing the effects of Zika virus infection early in gestation.



CONGENITAL MALFORMATIONS

Rubella – bilateral cataracts



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CONGENITAL MALFORMATIONS

HYPERTHERMIA

RADIATION – high levels of ionizing radiation

CHEMICAL
AGENTS/DRUGS

- Thalidomide - amelia or meromelia
- Nicotine and smoking - orofacial clefts, cardiac defects
- Alcohol - fetal alcohol syndrome (FAS); most common cause of mental retardation; binge drinking can cause FAE
- Cocaine- premature labor, abortion
- Dilantin, tetracycline, etc.
- Marijuana - neural tube defects
- Isotretinoin (Accutane)

CHEMICAL AGENTS/DRUGS

Thalidomide

- Amelia
- Meromelia
- Phocomelia



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FETAL ALCOHOL SPECTRUM DISORDER

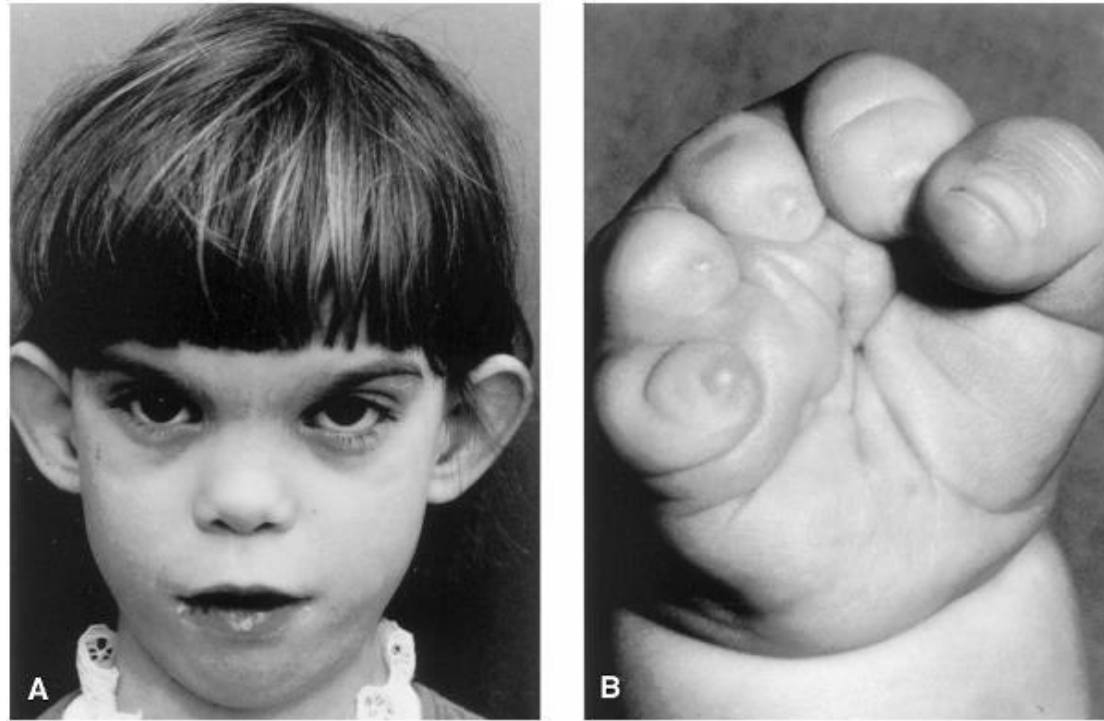
Fetal alcohol effects (FAE) – behavioral and learning disabilities; maternal alcohol abuse is now thought to be the most common cause of intellectual impairment.



Characteristic features of a child with fetal alcohol syndrome (FAS), including an indistinct philtrum, thin upper lip, depressed nasal bridge, short nose, and flat midface.

CHEMICAL AGENTS/DRUGS

- Dilantin



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CONGENITAL MALFORMATIONS

HORMONES

1. Androgens - ambiguous external genitalia
2. Diethylstilbestrol (DES) - adenocarcinoma of vagina; testicular malformations and abnormal sperm analysis
3. Cortisone - orofacial clefts
4. Birth control pills - appear to have low teratogenicity

MATERNAL DISEASE

Diabetes - high incidence of stillbirths, neonatal deaths, abnormally large infants, and congenital malformations

HYPOXIA - low birth weight infants

ENVIRONMENTAL CHEMICALS

organic mercury, mercury-containing fungicides

Lead

ADVANCED PATERNAL AGE - men transmit a higher number of mutations to their children than women

HORMONES

Congenital adrenal hyperplasia leads to ambiguous genitalia.



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2. Before we are born: Essentials of embryology and birth defects by Moore, K. L., Persaud, T. V. N., & Torchia, M. G., 9th ed., 2016, Elsevier